TEACHER RESOURCE GUIDE

FOR TEACHER USE ONLY

ADVANCED WATER SAFETY

Teacher Resource Guide | Year 9-10 | Victorian Curriculum F-10 Version 2.0

Lesson plans, answer key, A-E rubric, assessment checklist, differentiation guidance, and teacher background notes. Companion to the Year 9-10 Advanced Student Workbook.

STUDENT NAME

CLASS

TEACHER

Contents

1 Curriculum Alignment & Achievement Standards

Code	Content Description	Strand	How this unit addresses it
VCHPEP143	Investigate how health behaviours influence health outcomes for diverse groups	PSCH	Students investigate how gender, culture, and socioeconomic factors influence drowning rates
VCHPEP144	Evaluate health decisions and actions using health data	PSCH	Students evaluate prevention strategies against LSV/WHO evidence; calculate rates
VCHPEP147	Plan and evaluate strategies that build community capacity for safety	PSCH	Students design community interventions; audit school provision; propose policy actions
VCHPEP148	Evaluate health information and critically analyse community health strategies	PSCH	Students evaluate credibility of prevention evidence; compare international approaches
VCHPEM162	Apply specialised movement concepts in aquatic contexts	MPA	Survival swimming theory and rescue technique analysis (pool sessions if available)

Achievement Standard — Level 10 Key Points

- Critically analyse health information and the reliability of evidence in health contexts
- Plan and evaluate strategies that build individual and community capacity for health promotion
- Propose evidence-based community health solutions with measurable outcomes
- Apply and refine specialised movement skills; analyse performance using data

2 Unit Overview & Learning Intentions

Element	Detail
Unit Title	Advanced Water Safety: Data, Physiology & Prevention Policy
Year Level	Year 9-10 (Ages 14-16)
Duration	5 x 50-minute lessons (expandable with pool sessions)
Prerequisite	Year 7-8 Water Safety unit (flags, DRS ABCD, SwimSafe rules)
Assessment	Workbook (Sections 1-5) + Summative Task (Section 6: policy brief, campaign, or audit)
Key departure from Y7-8	Shifts from knowledge/application to analysis, evaluation, and synthesis

Essential Questions

- Why do drowning rates remain high despite awareness campaigns?
- Who is most at risk and why — and what does the evidence say about causes?
- What prevention strategies work best and why — and how do we know?
- What would you recommend to the Victorian Government to reduce drowning deaths?

3 Lesson Plans — 5 x 50 Minutes

Lesson 1 — Advanced Data Analysis — Victorian & Global Drowning Trends

50 min | VCHPEP143, VCHPEP144 | Classroom

ENGAGE (8 min)

"Victoria had 54 drowning deaths in 2023-24. Rate = 0.78 per 100,000. National rate = 1.2. Should we be proud of this, or worried?" Pair discussion then class share. Introduce: "At Year 9-10, describe is not enough. We analyse, evaluate, and justify."

EXPLORE (15 min)

Students complete Activity 1.1 calculations independently: percentage (Q1.1a), rate comparison (Q1.1b), CALD over-representation analysis (Q1.1c). Prompt: "Show your working. What does the number MEAN, not just what it is?"

EXPLAIN (12 min)

Teacher-led discussion: CALD over-representation. Introduce social determinants: access to swimming lessons, cultural norms, fishing risk, language barriers. Frame as systemic issue — not a commentary on any community. Reference: lsv.com.au/community/multicultural-safety/

ELABORATE (10 min)

Students complete data visualisation (Q1.1d): bar graph of deaths by location. Emphasise: labels, title, axes, interpretation — not just drawing bars.

EVALUATE (5 min)

Exit ticket: "In one sentence: what is the SINGLE most important data insight from today?"

Lesson 2 — Physiology of Drowning & CPR Mechanics

50 min | VCHPEP144, VCHPEM162 | Classroom

ENGAGE (8 min)

"CPR generates only 25-30% of normal cardiac output. Why is 25-30% enough to bridge survival?" Think-pair-share. Draw cardiac cycle vs CPR cycle on whiteboard.

EXPLORE (15 min)

Students complete Activity 2.1a (physiological sequence) and 2.1b (CPR physiology and 4-minute window). Key vocabulary: hypoxia, laryngospasm, cardiac arrest, cerebral hypoxia, cardiac output.

EXPLAIN (12 min)

Explain secondary drowning: rare but real. Symptoms hours after a near-drowning: coughing, fatigue, chest pain. Always seek medical review after any submersion incident. Walk through 1-10-1 cold water survival rule with Lake Eildon context.

ELABORATE (10 min)

Students complete Q2.1c (secondary drowning research — device required). Class discussion: "What is the implication for parents, teachers, coaches who supervise water activities?"

EVALUATE (5 min)

Exit ticket: "Explain the 4-minute window to someone who has never heard of it — in plain language."

Lesson 3 — Social Determinants of Drowning Risk

50 min | VCHPEP143, VCHPEP147 | Classroom

ENGAGE (8 min)

Values Clarification: Four Corners with statements like "Males drown more because they're less careful." Students defend their position. Point: these are over-simplifications. Introduce structural vs individual explanations.

EXPLORE (15 min)

Students complete Activity 3.1a (social determinants table) in pairs — assign different determinants to each pair for jigsaw reporting.

EXPLAIN (12 min)

Jigsaw sharing: each pair reports on their determinant (5 min). Teacher synthesises. Introduce WHO Social Determinants of Health Framework. Connect to Victorian data: CALD over-representation, male risk-taking, rural/inland waterway risk.

ELABORATE (12 min)

Students complete Q3.1b (3 school actions to reduce CALD disproportionality). Encourage specificity: WHERE? WHEN? WHO delivers them?

EVALUATE (3 min)

Exit: "Name ONE social determinant a school CAN address and ONE it CANNOT. Why is that distinction important?"

Lesson 4 — Prevention Strategies & International Comparison

50 min | VCHPEP144, VCHPEP148 | Classroom

ENGAGE (6 min)

"The government has $5 million for drowning prevention. 60 seconds — decide how to spend it." Quick-fire then reveal and discuss. This primes Activity 4.1.

EXPLORE (18 min)

Students complete Activity 4.1 (prevention strategy evaluation) using research and judgement. Students research Activity 5.1 (international comparison) using: royallifesaving.com.au, rlss.org.uk/education, redcross.org/take-a-class/water-safety.

EXPLAIN (10 min)

Class share: UK uses "Float to Live"; Red Cross uses "Think So You Don't Sink" framework; RLSSA uses the "Water Safety Framework." Highlight: what Australia does well (lifeguard system) vs areas for improvement (CALD engagement).

ELABORATE (12 min)

Students complete Q5.1a (identify one unadopted international element — argue for/against adoption). Prompt: "Evidence-based argument means citing the organisation, not just saying 'I think this is good.'"

EVALUATE (4 min)

Think-pair-share: "Which ONE prevention strategy has the strongest evidence? Which the weakest?"

Lesson 5 — Assessment Task — Planning, Drafting & Peer Review

50 min | VCHPEP147, VCHPEP148 | Classroom

ENGAGE (8 min)

Rapid revision Kahoot covering Lessons 1-4 content. Focus: rate calculations, physiological terms, social determinants, prevention strategy hierarchy.

ELABORATE (32 min)

Option A — Policy Brief: 600-800 words to Victorian Minister for Sport. Must cite 3+ sources with URLs; use Victorian data; address a social determinant; propose measurable outcomes. Scaffold: Situation -> Evidence -> Recommendations (x2) -> Success Metrics. Option B — Multimodal Campaign: Target ONE high-risk group from Victorian data. Must include: campaign name, key message, visual concept, target platform, 100-word justification. Option C — School Audit: Structured report benchmarked against Curriculum 2.0 (VCHPEP126, VCHPEM146) and LSV Classroom Toolkit. Include priority matrix.

EVALUATE (10 min)

Peer review using Assessment Checklist (Section 6 of this guide). Partners check: evidence present? Audience clear? Call to action specific? Sources cited with URLs?

4 Student Workbook — Answer Key

Answer Key Guidance:

Year 9-10 questions are largely analytical and evaluative — there is no single correct answer. Use these sample answers as quality BENCHMARKS for the A-level standard. Accept well-reasoned alternatives with supporting evidence.

Q1.1a — Male % calculation

Sample (A standard): 46 / 54 x 100 = 85.2%. This is higher than the global figure of ~80%, suggesting Victorian males take proportionally more risks near water than the global average male population. May reflect cultural norms, prevalence of unpatrolled inland waterways, influence of peer pressure and alcohol.

Q1.1b — Rate comparison

Calculation: 1.2 per 100,000 x 69 = 82.8 expected deaths at national average. Victoria (54) is actually BELOW the national average. Key insight: Victoria performs better overall but has specific problem areas (CALD, inland waterways). A student who calculates and recognises this nuance is at A standard.

Q1.1c — CALD over-representation

Sample (A standard): CALD Victorians are significantly over-represented — 40% of deaths vs ~28% of population. Over-representation ratio ~1.43:1. Two reasons beyond "language barriers": (1) Many CALD members grew up without access to swim lessons — skill gap is structural, not individual. (2) Higher participation in high-risk water activities (commercial fishing) with less safety training.

Q2.1a — Physiological sequence

Sample sequence (A standard): 1. Submersion/breath-holding/panic. 2. Laryngospasm (involuntary airway closure — "dry drowning" ~10-15% of cases). 3. In wet drowning: laryngospasm fails, water enters lungs, hypoxia begins. 4. Hypoxia -> loss of consciousness. 5. Cardiac arrest. 6. Cerebral hypoxia begins within 4-6 minutes — irreversible brain cell death. 7. Biological death. Students must use all five listed terms for full marks.

Q2.1b — Why early CPR matters

Sample (A standard): CPR generates ~25-30% of normal cardiac output. While insufficient for normal function, this delivers oxygenated blood to the brain at a rate that prevents irreversible cell death. After 4-6 minutes without oxygenated blood, neurons begin dying. After 10 minutes, neurologically intact survival probability drops dramatically. Bystander CPR within 4 minutes maintains minimum oxygenation until ROSC or defibrillation.

Q3.1a — Social determinants + prevention strategies

Accept any two well-reasoned determinants: gender norms, CALD background, SES, rural location, age. Prevention strategy must specifically target the mechanism — e.g. for masculinity norms: male-specific messaging that reframes water safety as strength (knowing when to walk away).

Q4.1a — $5M prevention justification

Sample (A standard): Priority 1 — Expand patrolled beach hours (strongest evidence: 0 deaths at patrolled sites). Priority 2 — Targeted CALD swimming access programs: subsidised lessons in-language by CALD community members (addresses structural access gap; has demonstrated impact in RLSSA pilot programs). Accept any well-justified combination with evidence from the prevention table.

Q5.1a — International element to adopt

Sample (A standard): UK RLSS "Float to Live" campaign explicitly teaches floating as the PRIMARY response to any water emergency — simpler and more physiologically sound than attempting to swim. Victorian messaging focuses primarily on swimming to safety. Floating requires no fitness, works in most water conditions, and directly counteracts panic-induced exhaustion. Adopting this as the core survival message would complement existing SwimSafe rules.

5 Assessment Rubric — A-E, 6 Criteria

This rubric applies to both the workbook and the summative task. Holistic judgement across criteria — not mechanical mark addition — produces the most valid grade.

Criterion	E — Emerging	D — Developing	C — Consolidating	B — Building	A — Extending
1. Data Analysis & Calculation VCHPEP144	Cannot perform basic calculations; misinterprets data	Calculations with errors; limited interpretation	Correctly calculates rates; reasonable interpretation	Accurate; interprets significance; identifies trends	Sophisticated calculations; nuanced interpretation; identifies implications and limitations
2. Physiological Knowledge VCHPEP144, VCHPEM162	Cannot describe drowning physiology; vocabulary absent	Names some processes; limited mechanism understanding	Describes drowning sequence accurately; uses most terms	Explains mechanisms precisely; connects physiology to CPR rationale	Sophisticated physiological analysis; all terms correct; explains clinical implications
3. Social Determinants VCHPEP143	Identifies individuals as responsible; no systemic analysis	Acknowledges social factors exist; limited analysis	Identifies key determinants; explains how they increase risk	Analyses multiple determinants with evidence; proposes targeted strategies	Sophisticated structural analysis; distinguishes individual vs systemic; evidence-based interventions
4. Prevention Evaluation VCHPEP147, VCHPEP148	Cannot distinguish stronger from weaker evidence	Names strategies; limited evaluation	Evaluates with reference to evidence; identifies limitations	Comparative evaluation using evidence hierarchy; justifies resource allocation	Critical evaluation; acknowledges trade-offs; sophisticated prioritisation with specific justification
5. International Comparison VCHPEP148	No research evident; cannot identify international approaches	Names one approach; superficial description	Compares at least two approaches against Victorian practice	Identifies substantive differences; evaluates relevance to Victorian context	Critical evidence-based comparison; argues for/against adoption; cites primary sources
6. Communication Product VCHPEP147	Incomplete; no clear audience, evidence, or call to action	Basic message; vague audience; weak citation	Clear message; appropriate audience; one data-backed claim with URL	Compelling; evidence-based; audience-appropriate; measurable call to action	Sophisticated, persuasive, evidence-rich; addresses barriers for target group; primary source citations

Report Comment Stems — Year 9-10 Water Safety

A standard: "[Name] demonstrated sophisticated analytical skills in interpreting Victorian and global drowning data, producing a highly persuasive and evidence-rich [product] that clearly addressed the needs of a specific at-risk group."

C standard: "[Name] showed sound knowledge of drowning prevention strategies and social determinants of health, producing a competent [product] that cited relevant data and identified a clear call to action."

E standard: "[Name] is developing their understanding of evidence-based health analysis. With additional support, [Name] will build confidence in using data to justify health recommendations."

6 Assessment Checklist — Summative Task

Use alongside the rubric. Share with students before submission — it functions as both a teacher marking guide and a student self-check.

All Options — Minimum Requirements

- [ ] At least 3 credible sources cited with full URLs
- [ ] At least 1 accurate Victorian or global statistic with source
- [ ] Clear identification of target audience or addressee
- [ ] A specific, measurable call to action or recommendation
- [ ] Evidence of drafting/planning in workbook planning space
- [ ] Connection to at least one social determinant or physiological concept from unit

Option A — Policy Brief

- [ ] Addressed to Victorian Minister for Sport (or equivalent)
- [ ] 600-800 words (or equivalent scope)
- [ ] Two specific recommendations clearly stated
- [ ] At least one counter-argument acknowledged and addressed
- [ ] Measurable success metric for each recommendation
- [ ] Victorian data cited (LSV 2023-24 or equivalent)
- [ ] References Victorian Curriculum or LSV/RLSSA framework

Option B — Multimodal Campaign

- [ ] ONE specific high-risk group identified from Victorian data
- [ ] Campaign name and key message clearly stated
- [ ] Visual concept described or shown
- [ ] Target platform specified
- [ ] 100-word justification of design decisions
- [ ] Addresses a specific barrier for the target group
- [ ] Evidence cited for the barrier addressed

Option C — School Audit

- [ ] Current provision described (what the school does now)
- [ ] Gaps identified with reference to Curriculum 2.0 codes
- [ ] Benchmarked against LSV Classroom Toolkit
- [ ] At least 3 specific recommendations with priority rationale
- [ ] Priority matrix included (urgency x impact or similar)
- [ ] Realistic implementation suggestions
- [ ] 400-word minimum (or equivalent scope)

7 Differentiation & Inclusion

Extension Students

- Option C (School Audit) is the most complex assessment task — recommend for highly capable students
- Encourage use of WHO and CDC international datasets for Section 1 calculations
- Research task: compare Victorian drowning prevention funding as % of health budget vs UK/USA
- Invite guest speaker: LSV community engagement officer, RLSSA researcher, or local council water safety coordinator

Students Requiring Support

- Provide the calculation scaffold for Section 1: rate = deaths / population units x 100,000
- Allow verbal responses for physiological sequence questions (Q2.1a, Q2.1b)
- For Section 3 social determinants: provide a pre-populated list — students choose two and explain
- Option B (multimodal campaign) is the most accessible assessment task
- Pair with a peer research partner for the international comparison activity

CALD Students

- The CALD over-representation data may be personally significant — frame as systemic, not individual
- Option C (School Audit) may surface valuable community knowledge — CALD experience is data too
- LSV translated materials can be referenced: lsv.com.au/community/multicultural-safety/
- For Option B: encourage targeting their own community — most impactful application of unit

EAL/D Students

- Provide vocabulary glossary: hypoxia, laryngospasm, cardiac output, determinants, cerebral
- Allow diagram/annotated visual as alternative to written physiological sequence (Q2.1a)
- Sentence starters for policy brief: "The evidence suggests...", "Specifically, I recommend..."
- Assessment task: allow mixed language for planning notes; final product in English

8 Teacher Background Notes

Physiology Quick Reference

- **Hypoxia:** Oxygen deprivation at tissue level. Cerebral hypoxia begins within 4 minutes of cardiac arrest.
- **Laryngospasm:** Involuntary closure of vocal cords on water contact. "Dry drowning" ~10-15% of cases.
- **CPR cardiac output:** ~25-30% of normal resting output — enough to prevent brain death for 10-15 minutes pending defibrillation.
- **Cold water shock:** Below ~15 degrees C triggers involuntary gasp, hyperventilation, vasoconstriction, cardiac arrhythmia risk.
- **Secondary drowning:** Pulmonary oedema developing hours after near-drowning. Requires medical review.

Curriculum 2.0 Implementation Note

At Year 9-10, the critical move is from understanding (can explain what and why) to evaluation and synthesis (can judge quality of evidence and produce new health products). Sections 1-3 build analytical vocabulary; Sections 4-5 require judgement; Section 6 demands synthesis.

9 Resources & Sources

Organisation	URL
Life Saving Victoria — Drowning Reports	lsv.com.au/research/victorian-drowning-reports/
LSV Classroom Toolkit	lsv.com.au/toolkit/classroom_based_water_safety.php
LSV Multicultural Safety	lsv.com.au/community/multicultural-safety/
Royal Life Saving Society Australia	royallifesaving.com.au
WHO Drowning Fact Sheet	who.int/news-room/fact-sheets/detail/drowning
RLSS UK Education Resources	rlss.org.uk/education
American Red Cross Water Safety	redcross.org/take-a-class/water-safety
Victorian Curriculum (VCAA)	victoriancurriculum.vcaa.vic.edu.au
Paddle Australia	paddle.org.au
CDC Drowning Prevention	cdc.gov/drowning/prevention